

Solutions to Aluffi's Algebra: Chapter 0

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VI: Linear Algebra

§1 Free modules revisited

1.7

Let R be an integral domain, and let $M = R^{\oplus A}$ be a free R -module. Let K be the field of fractions of R , and view M as a subset of $V = K^{\oplus A}$ in the evident way. Prove that a subset $S \subseteq M$ is linearly independent in M (over R) if and only if it is linearly independent in V (over K). Conclude that the rank of M (as an R -module) equals the dimension of V (as a K -vector space). Prove that if S generates M over R , then it generates V over K . Is the converse true?

Solution.

Let $R \hookrightarrow K$ via $r \mapsto \frac{r}{1}$. This induces an inclusion $R^{\oplus A} \hookrightarrow K^{\oplus A}$ defined by:

$$\sum r_i a_i \mapsto \sum \frac{r_i}{1} a_i$$

$\implies$ Assume $S \subset M$ is linearly independent over R . Suppose we have a linear relation in V :

$$\sum_{i \in I} \frac{r_i}{p_i} s_i = 0$$

where $|I| < +\infty$. Then:

$$\sum_{i \in I} P \frac{r_i}{p_i} s_i = 0, \quad \text{where } P = \prod_{i \in I} p_i$$

For all $k \in I$, the term $\frac{P}{p_k} = \prod_{i \in I \setminus \{k\}} p_i$ is an element of R . Hence, we have obtained a linear relation in M . Since S is linearly independent in M , all coefficients must be zero, which implies $r_i = 0$ for all $i \in I$.

$\longleftarrow$ Obvious.

Suppose that S generates M over R and let $v \in V$. Then we can write v as a finite sum:

$$v = \sum_{i \in I} \frac{r_i}{p_i} a_i$$

Let $P = \prod_{i \in I} p_i$. Multiplying by P clears the denominators, meaning $Pv \in M$. Since S generates M , there exist $r_j \in R$ and $s_j \in S$ such that:

$$Pv = \sum_{j \in J} r_j s_j$$

Dividing by P in V , we get:

$$v = \sum_{j \in J} \frac{r_j}{P} s_j \in \langle S \rangle_K$$

Thus, S generates V over K . The converse is not true. For example, the subset $\{2\}$ generates $\mathbb{Q}$ over $\mathbb{Q}$, but does not generate $\mathbb{Z}$ over $\mathbb{Z}$. $\square$

1.9

Let R be a commutative ring, and let M be an R -module. Let $\mathfrak{m}$ be a maximal ideal in R , such that $\mathfrak{m}M = 0$ (that is, $rm = 0$ for all $r \in \mathfrak{m}$, $m \in M$). Define in a natural way a vector space structure over $R/\mathfrak{m}$ on M .

Solution.

We define scalar multiplication by $\bar{r}m := rm$. We must check that this operation is correctly defined. If $\bar{r} = \bar{r}'$, then:

$$r - r' \in \mathfrak{m} \implies (r - r')m = 0 \implies rm = r'm$$

Thus, the operation is well-defined. The other vector space axioms trivially hold because M is an R -module. $\square$

1.10

Let R be a commutative ring, and let F be a free R -module with basis B . Let $\mathfrak{m}$ be a maximal ideal of R , and let $k = R/\mathfrak{m}$. Prove that $F/\mathfrak{m}F \cong k^{\oplus B}$ as k -vector spaces.

Solution.

Let $\sum r_i b_i \in F$. If $x \in \mathfrak{m}F$, then $x = \sum m_i f_i = \sum m_i \sum r_{ij} b_j = \sum m'_i b_i$, where $m'_i \in \mathfrak{m}$. On the other hand, all $\sum m_i b_i$ with $m_i \in \mathfrak{m}$ are in $\mathfrak{m}F$. Hence, $\mathfrak{m}F = \mathfrak{m}^{\oplus B}$.

$F/\mathfrak{m}F$ is an R -module, and for all $m \in \mathfrak{m}$ and for all $\bar{f} \in F/\mathfrak{m}F$:

$$m\bar{f} = \overline{mf} = \bar{0}$$

Hence, by Exercise 1.9, $F/\mathfrak{m}F$ is a k -vector space.

Now let's prove that $\pi(B)$ is a basis of $F/\mathfrak{m}F$, where π is the natural projection. Obviously, $\pi(B)$ generates $F/\mathfrak{m}F$ since B generates F . Suppose that $\sum_I (r_i + \mathfrak{m})(b_i + \mathfrak{m}F) = 0 + \mathfrak{m}F$. Then:

$$\left(\sum r_i b_i\right) + \mathfrak{m}F = \bar{0} \implies \sum r_i b_i \in \mathfrak{m}F \implies r_i \in \mathfrak{m} \implies r_i + \mathfrak{m} = \bar{0}$$

This implies that $\pi(B)$ is linearly independent, hence it is a basis. Therefore, $F/\mathfrak{m}F \cong k^{\oplus B}$. $\square$

1.11

Prove that commutative rings satisfy the IBN (Invariant Basis Number) property.

Solution.

Let R be a commutative ring which has a proper ideal and let F be a free R -module. Suppose $F \simeq R^{\oplus A} \simeq R^{\oplus B}$.

R has a maximal ideal $\mathfrak{m}$ and by Exercise 1.10,

$$k^{\oplus A} \simeq F/\mathfrak{m}F \simeq k^{\oplus B}$$

where $k = R/\mathfrak{m}$. But then it follows that $A \simeq B$, since k is a field. $\square$

1.13

Let A be an abelian group such that $\text{End}_{\text{Ab}}(A)$ is a field of characteristic 0. Prove that $A \cong \mathbb{Q}$. (Hint: Prove that A carries a $\mathbb{Q}$ -vector space structure; what must its dimension be?)

Solution.

$\text{End}(A)$ acts on A : $\varphi a := \varphi(a)$ and makes the latter into a vector space.

We have an inclusion $\mathbb{Q} \hookrightarrow \text{End}(A)$, since $\text{char}(\text{End } A) = 0$:

$$\frac{p}{q} \mapsto (p \text{ id}) \circ (q \text{ id})^{-1}$$

Hence, A is a $\mathbb{Q}$ -vector space via

$$\frac{p}{q}a = ((p \text{ id}) \circ (q \text{ id})^{-1})(a) = (q \text{ id})^{-1}(pa)$$

Basically, $p \cdot a = (p \text{ id})(a) = pa$ for integer p , and $\frac{1}{q}a = b$ such that $a = qb$. It exists and is unique because $q \text{ id}$ has an inverse, hence is an isomorphism.

Now, suppose $f \in \text{End}_{\text{Ab}}(A)$. Then

$$\forall n \in \mathbb{Z} \quad f(na) = nf(a)$$

We have also $f(a) = mf\left(\frac{1}{m}a\right) \implies f\left(\frac{1}{m}a\right) = \frac{1}{m}f(a)$ for $m \in \mathbb{N}$.

Hence, $f\left(\frac{n}{m}a\right) = nf\left(\frac{1}{m}a\right) = \frac{n}{m}f(a)$ or simply

$$f(ra) = rf(a) \quad \forall r \in \mathbb{Q}$$

This means that f is actually a $\mathbb{Q}$ -vector space endomorphism, so $\text{End}_{\text{Ab}}(A) = \text{End}_{\mathbb{Q}\text{-Vect}}(A)$.

Suppose S is a basis of A , $|S| > 1$ and $a \in S$. Then $A = V \oplus W$, where $V = \langle a \rangle$ and $W = \langle S \setminus \{a\} \rangle$. Define a map $\sigma : A \rightarrow A$ via

$$A \simeq V \oplus W \xrightarrow{\pi_1} V \hookrightarrow A$$

which sends $\lambda a + \sum_{s \in S \setminus \{a\}} \lambda_s s$ to λa .

It is a $\mathbb{Q}$ -vector space homomorphism as a composition of homomorphisms and has a non-trivial kernel, since $S \setminus \{a\} \subset \ker \sigma$ and $S \setminus \{a\}$ is non-empty and does not contain 0. Hence, σ does not have an inverse, contradicting the fact that $\text{End}(A)$ is a field.

It follows that $|S| = \dim A \leq 1$. The case of 0 is not possible, so $\dim_{\mathbb{Q}} A = 1$ and $A \simeq \mathbb{Q}$.

We can also show that for a $\mathbb{Q}$ -vector space V , $\text{End}_{\mathbb{Q}\text{-Vect}}(V) \simeq M_n(\mathbb{Q})$ as rings, where $n = \dim V$. Thus, in the general case, $\text{End}_{\text{Ab}}(V)$ is not a field, and $\mathbb{Q}$ is the only space with its endomorphism ring being a field. $\square$

1.19

Let k be a field, and let $f(x) \in k[x]$ be any polynomial. Prove that there exists a multiple of $f(x)$ in which all exponents of nonzero monomials are *prime* integers. (Example: for $f(x) = 1 + x^5 + x^6$, $(1 + x^5 + x^6)(2x^2 - x^3 + x^5 - x^8 + x^9 - x^{10} + x^{11}) = 2x^2 - x^3 + x^5 + 2x^7 + 2x^{11} - x^{13} + x^{17}$.)

Solution.

Let $V = k[x]/(f(x))$. This is a k -vector space with basis $\overline{1}, \overline{x}, \dots, \overline{x^{n-1}}$, where $n = \deg f$.

Take $n + 1$ distinct primes $p_1, \dots, p_{n+1}$. The system

$$S = \{\overline{x^{p_1}}, \overline{x^{p_2}}, \dots, \overline{x^{p_{n+1}}}\}$$

is linearly dependent, since $|S| = n + 1 > \dim V$. So, for some $\lambda_i \in k$ (not all zero):

$$\sum_{i=1}^{n+1} \lambda_i \overline{x^{p_i}} = \overline{0} \implies \sum_{i=1}^{n+1} \lambda_i x^{p_i} \in (f(x))$$

$$\implies \exists p(x) \in k[x] \quad \text{such that} \quad f(x)p(x) = \sum_{i=1}^{n+1} \lambda_i x^{p_i}$$

This multiple has only prime exponents, as desired. $\square$

§2 Homomorphisms of free modules, I

§3 Homomorphisms of free modules, II

Let R be a commutative ring and $M = \langle m_1, \dots, m_r \rangle$ a finitely generated R -module. Let $A \in M_r(R)$ be a matrix such that

$$A \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Prove that $\det(A)m = 0$ for all $m \in M$.

Solution.

Let A^* be the adjoint matrix of A . Recall the fundamental property $A^*A = \det(A)I_r$. Starting from the given equation:

$$A \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Multiply both sides on the left by A^* :

$$A^*A \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = A^* \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\det(A)I_r \begin{pmatrix} m_1 \\ \vdots \\ m_r \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

This implies $\det(A)m_i = 0$ for every generator m_i . Since any $m \in M$ is an R -linear combination of these generators, $\det(A)m = 0$ for all $m \in M$. $\square$

3.7

Let R be a commutative ring, M a finitely generated R -module, and let J be an ideal of R . Assume $JM = M$. Prove that there exists an element $b \in J$ such that $(1 + b)M = 0$.

Solution.

Since $M = JM$, each generator m_i can be written as $m_i = \sum_{j=1}^r b_{ij}m_j$ with $b_{ij} \in J$. In matrix form, this is $\mathbf{m}^T = B\mathbf{m}^T$, or $(I - B)\mathbf{m}^T = 0$. By Exercise 3.6, $\det(I - B)$ annihilates M .

Expanding the determinant $\det(I - B)$ modulo J , we see that since all entries of B are in J :

$$\det(I - B) \equiv \det(I) \equiv 1 \pmod{J}$$

Thus, $\det(I - B) = 1 + b$ for some $b \in J$. Hence $(1 + b)M = 0$. $\square$

3.8

Let R be a commutative ring, M be a finitely generated R -module, and let J be an ideal of R contained in the Jacobson radical of R (Exercise V.3.14). Prove that $M = 0 \iff JM = M$.

Solution.

$\implies$ If $M = 0$, then $JM = 0 = M$.

$\impliedby$ Assume $JM = M$. By Exercise 3.7, there exists $b \in J$ such that $(1 + b)M = 0$. Since J is contained in the Jacobson radical, $1 + b$ is a unit in R by Exercise V.3.14. Hence, $M = (1 + b)^{-1}(1 + b)M = 0$. $\square$

3.9

Let $(R, \mathfrak{m})$ be a commutative local ring (meaning a ring with a single maximal ideal $\mathfrak{m}$) and M, N be finitely generated R -modules. Prove that if $M = \mathfrak{m}M + N$, then $M = N$.

Solution.

Let $L = M/N$. Since M is a finitely generated R -module, its quotient is also finitely generated. For any ideal $I \subseteq R$ and submodule $N \subseteq M$, we have

$$I(M/N) = \frac{IM + N}{N}$$

Applying this to the maximal ideal $\mathfrak{m}$, we have:

$$\mathfrak{m}(M/N) = \frac{\mathfrak{m}M + N}{N} = M/N$$

In a local ring, the Jacobson radical $J(R)$ is equal to the unique maximal ideal $\mathfrak{m}$. Therefore, we have $J(R)(M/N) = M/N$. Since M/N is finitely generated, Nakayama's Lemma implies that $M/N = 0$. Hence, $M = N$. $\square$

3.16

Every complex $V_\bullet$ determines an element in $K(k\text{-Vect}^f)$ (Grothendieck group), namely

$$\chi_K(V_\bullet) := \sum_i (-1)^i [V_i] \in K(k\text{-Vect}^f).$$

- χ_K 'is an Euler characteristic', in the sense that it satisfies the formula given in Proposition 3.13: $\chi_K(V_\bullet) = \sum_i (-1)^i [H_i(V_\bullet)]$.
- χ_K is a 'universal Euler characteristic', in the following sense. Let G be an abelian group, and let δ be a function associating an element of G to each finite-dimensional vector space, such that $\delta(V) = \delta(V')$ if $V \cong V'$ and $\delta(V/U) = \delta(V) - \delta(U)$. For $V_\bullet$ a complex, define $\chi_G(V_\bullet) = \sum_i (-1)^i \delta(V_i)$. Then δ induces a (unique) group homomorphism $K(k\text{-Vect}^f) \rightarrow G$ mapping $\chi_K(V_\bullet)$ to $\chi_G(V_\bullet)$.

- In particular, $\delta = \dim$ induces a group homomorphism $K(k\text{-Vect}^f) \rightarrow \mathbb{Z}$ such that $\chi_K(V_\bullet) \mapsto \chi(V_\bullet)$.
- This is in fact an isomorphism.

Solution.

Part 1: χ_K is an Euler characteristic. We prove the formula $\chi_K(V_\bullet) = \sum_i (-1)^i [H_i(V_\bullet)]$ by induction on the length n of the complex $V_\bullet$.

Base case $n = 0$: Consider the complex $0 \rightarrow V_0 \rightarrow 0$. We have $H_0(V_\bullet) = V_0$. Thus:

$$\chi_K(V_\bullet) = [V_0] = [H_0(V_\bullet)]$$

Base case $n = 1$: Consider $0 \rightarrow V_1 \xrightarrow{\alpha} V_0 \rightarrow 0$. The homologies are $H_1(V_\bullet) = \ker \alpha$ and $H_0(V_\bullet) = V_0 / \operatorname{im} \alpha$. We have the short exact sequences:

$$0 \rightarrow \ker \alpha \rightarrow V_1 \rightarrow \operatorname{im} \alpha \rightarrow 0 \implies [V_1] = [\ker \alpha] + [\operatorname{im} \alpha]$$

$$0 \rightarrow \operatorname{im} \alpha \rightarrow V_0 \rightarrow V_0 / \operatorname{im} \alpha \rightarrow 0 \implies [V_0] = [\operatorname{im} \alpha] + [V_0 / \operatorname{im} \alpha]$$

Computing the Euler characteristic:

$$\begin{aligned} \chi_K(V_\bullet) &= [V_0] - [V_1] = ([\operatorname{im} \alpha] + [V_0 / \operatorname{im} \alpha]) - ([\ker \alpha] + [\operatorname{im} \alpha]) = [V_0 / \operatorname{im} \alpha] - [\ker \alpha] \\ &= [H_0(V_\bullet)] - [H_1(V_\bullet)] = \sum_{i=0}^1 (-1)^i [H_i(V_\bullet)] \end{aligned}$$

Inductive step: Assume the statement holds for complexes of length n . Consider a complex $V_\bullet$ of length $n + 1$:

$$0 \rightarrow V_{n+1} \xrightarrow{\alpha_{n+1}} V_n \xrightarrow{\alpha_n} \dots \xrightarrow{\alpha_1} V_0 \rightarrow 0$$

Let $V'_\bullet$ be its truncated complex $0 \rightarrow V_n \xrightarrow{\alpha_n} \dots \rightarrow V_0 \rightarrow 0$. By the induction hypothesis, $\chi_K(V'_\bullet) = \sum_{i=0}^n (-1)^i [H_i(V'_\bullet)]$. By definition, $\chi_K(V_\bullet) = \chi_K(V'_\bullet) + (-1)^{n+1} [V_{n+1}]$. Let us compare the homologies of $V_\bullet$ and $V'_\bullet$:

- $H_i(V_\bullet) = H_i(V'_\bullet)$ for all $i < n$.
- $H_n(V'_\bullet) = \ker \alpha_n$.
- $H_n(V_\bullet) = \ker \alpha_n / \operatorname{im} \alpha_{n+1}$.
- $H_{n+1}(V_\bullet) = \ker \alpha_{n+1}$.

The exact sequence $0 \rightarrow \ker \alpha_{n+1} \rightarrow V_{n+1} \rightarrow \operatorname{im} \alpha_{n+1} \rightarrow 0$ gives $[V_{n+1}] = [\ker \alpha_{n+1}] + [\operatorname{im} \alpha_{n+1}]$. The exact sequence $0 \rightarrow \operatorname{im} \alpha_{n+1} \rightarrow \ker \alpha_n \rightarrow \ker \alpha_n / \operatorname{im} \alpha_{n+1} \rightarrow 0$ gives $[\ker \alpha_n] = [\operatorname{im} \alpha_{n+1}] + [\ker \alpha_n / \operatorname{im} \alpha_{n+1}]$, which rewrites as $[H_n(V'_\bullet)] = [\operatorname{im} \alpha_{n+1}] + [H_n(V_\bullet)]$.

Now, sum the alternating homologies for $V_\bullet$:

$$\sum_{i=0}^{n+1} (-1)^i [H_i(V_\bullet)] = \sum_{i=0}^{n-1} (-1)^i [H_i(V'_\bullet)] + (-1)^n [H_n(V_\bullet)] + (-1)^{n+1} [H_{n+1}(V_\bullet)]$$

Substitute $[H_n(V_\bullet)] = [H_n(V'_\bullet)] - [\text{im } \alpha_{n+1}]$ and $[H_{n+1}(V_\bullet)] = [\ker \alpha_{n+1}]$:

$$\begin{aligned} &= \sum_{i=0}^{n-1} (-1)^i [H_i(V'_\bullet)] + (-1)^n ([H_n(V'_\bullet)] - [\text{im } \alpha_{n+1}]) + (-1)^{n+1} [\ker \alpha_{n+1}] \\ &= \sum_{i=0}^n (-1)^i [H_i(V'_\bullet)] + (-1)^{n+1} ([\text{im } \alpha_{n+1}] + [\ker \alpha_{n+1}]) \\ &= \chi_K(V'_\bullet) + (-1)^{n+1} [V_{n+1}] = \chi_K(V_\bullet) \end{aligned}$$

This completes the induction.

Part 2: χ_K is a universal Euler characteristic. Recall that $K(k\text{-Vect}^f) = F/E$, where F is the free abelian group on isomorphism classes of finite-dimensional vector spaces, and E is the subgroup generated by relations from short exact sequences. Define a set-map from the generators of F to G via $[V] \mapsto \delta(V)$. Because $\delta(V) = \delta(V')$ whenever $V \cong V'$, this map is correctly defined and extends linearly to a unique group homomorphism $\varphi : F \rightarrow G$.

If $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$ is exact, then $W \cong V/U$. Thus:

$$\varphi([V] - [U] - [W]) = \delta(V) - \delta(U) - \delta(V/U) = 0$$

Hence, the subgroup $E \subset \ker \varphi$. By the universal property of quotients, φ factors through to a unique induced morphism $\varphi^* : K(k\text{-Vect}^f) \rightarrow G$ such that $\varphi^*([V]) = \delta(V)$. Applying this to a complex $V_\bullet$:

$$\varphi^*(\chi_K(V_\bullet)) = \varphi^*\left(\sum_i (-1)^i [V_i]\right) = \sum_i (-1)^i \varphi^*([V_i]) = \sum_i (-1)^i \delta(V_i) = \chi_G(V_\bullet)$$

To see φ^* is unique, suppose $\sigma : K(k\text{-Vect}^f) \rightarrow G$ also maps $\chi_K(V_\bullet)$ to $\chi_G(V_\bullet)$ for all complexes. Consider the trivial complex $0 \rightarrow V \rightarrow 0$. Here $\chi_K(V_\bullet) = [V]$ and $\chi_G(V_\bullet) = \delta(V)$. This forces $\sigma([V]) = \delta(V)$, so $\sigma = \varphi^*$.

Part 3 & 4: $\delta = \dim$ induces an isomorphism to $\mathbb{Z}$. Since $\dim(V/U) = \dim V - \dim U$ and dimension is an isomorphism invariant, we apply Part 2. The function $\delta = \dim$ induces a group homomorphism $\varphi : K(k\text{-Vect}^f) \rightarrow \mathbb{Z}$ mapping $\chi_K(V_\bullet) \mapsto \chi(V_\bullet)$. We show φ is an isomorphism.

First, observe a key property in $K(k\text{-Vect}^f)$. For any spaces V, U , there is an exact sequence

$$0 \rightarrow U \rightarrow V \oplus U \rightarrow V \rightarrow 0$$

It follows that $[V \oplus U] = [V] + [U]$. For any n -dimensional vector space V , $V \cong k^{\oplus n}$. Therefore, $[V] = n[k] = (\dim V)[k]$.

Surjectivity: The morphism φ maps $[V] \mapsto \dim V$. It is clearly surjective onto $\mathbb{Z}$, as choosing $V = k^{\oplus n}$ yields n .

Injectivity: Let $x \in K(k\text{-Vect}^f)$ be any element. We can write x as a finite linear combination $\sum \lambda_i [V_i]$ for some $\lambda_i \in \mathbb{Z}$. Suppose $x \in \ker \varphi$. Then:

$$0 = \varphi\left(\sum \lambda_i [V_i]\right) = \sum \lambda_i \varphi([V_i]) = \sum \lambda_i \dim V_i$$

Using the identity $[V_i] = (\dim V_i)[k]$ derived above, we rewrite our element x in K :

$$\sum \lambda_i [V_i] = \sum \lambda_i (\dim V_i \cdot [k]) = \left(\sum \lambda_i \dim V_i \right) [k] = 0 \cdot [k] = 0$$

Thus, $\ker \varphi = \{0\}$, meaning φ is injective. Therefore, φ is an isomorphism. $\square$

§4 Presentations and resolutions

4.3

Prove that an integral domain R is a PID if and only if every submodule of R itself is free.

Solution.

$\implies$ Let $I \subset R$ be a submodule. By definition, submodules of R are exactly the ideals of R . Since R is a PID, $I = (a)$ for some $a \in R$. If $I = 0$, it is free (with an empty basis). If $I \neq 0$, the map $R \rightarrow I$ given by $r \mapsto ra$ is an isomorphism of R -modules. Alternatively, the set $\{a\}$ is linearly independent since R is an integral domain ($ra = 0 \implies r = 0$ for $a \neq 0$), and it generates I . Thus, it is a basis of I , which implies $I \simeq R^1$.

$\Leftarrow$ Suppose $I \subset R$ is an ideal. By assumption, it is free as an R -module. Every basis of I would be a linearly independent system in R . However, any two non-zero elements $x, y \in R$ are linearly dependent over R , since $y \cdot x - x \cdot y = 0$. Therefore, the maximum size of any linearly independent set in R is 1.

This means $\text{rk } I \leq 1$. Hence, the basis is either empty ($I = 0$) or consists of a single element ($I = (a)$ for some $a \in R$). Since every ideal is generated by at most one element, R is a PID. $\square$

4.4

Let R be a commutative ring and M an R -module.

- Prove that $\text{Ann}(M)$ is an ideal of R .
- If R is an integral domain and M is finitely generated, prove that M is torsion if and only if $\text{Ann}(M) \neq 0$.
- Give an example of a torsion module M over an integral domain, such that $\text{Ann}(M) = 0$. (Of course this example cannot be finitely generated!)

Solution.

$$\text{Ann}(M) = \{r \in R \mid rm = 0 \quad \forall m \in M\}$$

$$a \in R, r \in \text{Ann}(M) \implies \forall m \in M, \quad (ar)m = a(rm) = a0 = 0.$$

$$\forall r, r' \in \text{Ann}(M), \forall m \in M, \quad (r + r')m = rm + r'm = 0 + 0 = 0.$$

More is true: $\forall m \in M$ $\text{Ann}(m)$ is an ideal of R and $\text{Ann}(M) = \bigcap_{m \in M} \text{Ann}(m)$

Note that if R is an integral domain and M is finitely generated, there is a surjective morphism $R^{\oplus A} \rightarrow M$, $A = \{a_1, \dots, a_n\}$, hence $M \simeq R^{\oplus A}/N$ for some submodule N .

$\Leftarrow$ If $\text{Ann}(M) \neq 0$, then $\exists r \in R \mid r \neq 0, \quad rm = 0 \quad \forall m \in M \implies \forall m \in M, \quad m$ is torsion.

$\implies$ If M is torsion, then $\exists r_i \in R, \quad i = 1, \dots, n \mid r_i \bar{a}_i = \bar{0}. \implies \forall m \in M, \quad m = \sum p_i \bar{a}_i \implies rm = 0$, where $r = \prod r_i \neq 0 \implies \text{Ann}(M) \neq 0$.

$\mathbb{Q}$ is not finitely generated as a $\mathbb{Z}$ -module. $\mathbb{Z} \subset \mathbb{Q}$ is a submodule. $\mathbb{Q}/\mathbb{Z}$ as a $\mathbb{Z}$ -module is torsion:

$$q \left(\frac{p}{q} + \mathbb{Z} \right) = p + \mathbb{Z} = \mathbb{Z}$$

If $n \in \mathbb{Z}$ such that $\forall \bar{s} \in \mathbb{Q}/\mathbb{Z}, \quad n\bar{s} = \bar{0}$, then $\forall m \in \mathbb{N}, \quad n \left(\frac{1}{m} + \mathbb{Z} \right) = \frac{n}{m} + \mathbb{Z} = \mathbb{Z} \implies m \mid n \quad \forall m \in \mathbb{N} \implies n = 0$.

Hence, $\text{Ann}_{\mathbb{Z}}(\mathbb{Q}/\mathbb{Z}) = 0$. □

4.5

Let M be a module over a commutative ring R . Prove that an ideal I of R is the annihilator of an element of M if and only if M contains an isomorphic copy of R/I (viewed as an R -module).

The *associated primes* of M are the prime ideals among the ideals $\text{Ann}(m)$, for $m \in M$. The set of the associated primes of a module M is denoted $\text{Ass}_R(M)$. Note that every prime in $\text{Ass}_R(M)$ contains $\text{Ann}_R(M)$.

Solution.

$\implies$ If $I = \text{Ann}(m)$ is an ideal, then $\ker \varphi = I$, where $\varphi : R \rightarrow M$ defined by $\varphi(r) = rm$ is an R -module homomorphism. Hence, we have an inclusion $R/I \simeq \text{im } \varphi \hookrightarrow M$.

$\Leftarrow$ If there is an inclusion $\varphi : R/I \hookrightarrow M$, then consider $m = \varphi(\bar{1})$. If $rm = 0$, then

$$0 = r\varphi(\bar{1}) = \varphi(r\bar{1}) = \varphi(\bar{r}) \implies \bar{r} = \bar{0} \quad (\text{since } \ker \varphi = 0) \implies r \in I$$

This argument works in the reverse direction as well, so $I = \text{Ann}(m)$.

Now, suppose that $\text{Ann}(m)$ is an associated prime. We need to show that $\text{Ann}(m) \supset \text{Ann}(M)$. It is obvious, since

$$\text{Ann}(M) = \bigcap_{m \in M} \text{Ann}(m)$$

□

4.6

Let M be a module over a commutative ring R , and consider the family of ideals $\text{Ann}(m)$, as m ranges over the nonzero elements of M . Prove that the maximal elements in this family are prime ideals of R . Conclude that if R is Noetherian, then $\text{Ass}_R(M) \neq \emptyset$.

Solution.

Note that $\forall r \in R, \quad \text{Ann}(m) \subset \text{Ann}(rm)$, for if $sm = 0$, then $s(rm) = r(sm) = 0$.

Suppose $\text{Ann}(m)$ is maximal among annihilators and that $rs \in \text{Ann}(m)$. If $r \notin \text{Ann}(m)$, then $rm \neq 0$.

$s(rm) = 0 \implies s \in \text{Ann}(rm)$. Since $\text{Ann}(m)$ is maximal and $\text{Ann}(m) \subset \text{Ann}(rm)$ (with $rm \neq 0$), we must have $\text{Ann}(m) = \text{Ann}(rm)$. Hence, $s \in \text{Ann}(m)$ and $\text{Ann}(m)$ is prime.

The family of ideals $\mathcal{F} = \{\text{Ann}(m) \mid m \in M \setminus \{0\}\}$ is non-empty.

In a Noetherian ring, every non-empty family of ideals has a maximal element with respect to inclusion. Let P be a maximal element of $\mathcal{F}$. By the argument above, P is a prime ideal. Because $P \in \mathcal{F}$, $P = \text{Ann}(m)$ for some non-zero $m \in M$, meaning P is an associated prime of M .

Therefore, $\text{Ass}_R(M) \neq \emptyset$. □

4.13

Let R be a commutative ring. A tuple $(a_1, a_2, \dots, a_n)$ of elements of R is a *regular sequence* if a_1 is a non-zero-divisor in R , a_2 is a non-zero-divisor modulo (a_1) , a_3 is a non-zero-divisor modulo (a_1, a_2) , and so on.

For a, b in R , consider the following complex of R -modules:

$$(*) \quad 0 \rightarrow R \xrightarrow{d_2} R \oplus R \xrightarrow{d_1} R \xrightarrow{\pi} \frac{R}{(a, b)} \rightarrow 0$$

where π is the canonical projection, $d_1(r, s) = ra + sb$, and $d_2(t) = (bt, -at)$. Put otherwise, d_1 and d_2 correspond, respectively, to the matrices

$$\begin{pmatrix} a & b \end{pmatrix}, \quad \begin{pmatrix} b \\ -a \end{pmatrix}.$$

- Prove that this is indeed a complex, for every a and b .
- Prove that if (a, b) is a regular sequence, this complex is *exact*.

The complex $(*)$ is called the *Koszul complex* of (a, b) . Thus, when (a, b) is a regular sequence, the Koszul complex provides us with a free resolution of the module $R/(a, b)$. [4.14, 5.4, VIII.4.22]

Solution.

$$[d_1 \circ d_2] = (a \ b) \begin{pmatrix} b \\ -a \end{pmatrix} = 0, \quad \text{so } d_1 \circ d_2 = 0$$

$\pi \circ d_1 = 0$ is obvious, and canonical projection is an epimorphism.

If (a, b) is regular and $d_2(t) = 0$, then

$$(bt, -at) = (0, 0) \implies at = 0 \implies t = 0$$

since a is a non-zero-divisor. Hence, $\ker d_2 = 0$.

If $d_1(r, s) = ra + sb = 0$, then consider this equation modulo (a) :

$$\bar{s}\bar{b} = \bar{0} \implies \bar{s} = \bar{0}$$

for $\bar{b}$ is a NZD (non-zero-divisor).

$$\implies s = at \quad \text{for some } t \in R.$$

Then

$$0 = ra + atb = a(r + tb) \implies r = -tb$$

Thus, $(r, s) = (-bt, at) = d_2(-t) \in \text{im } d_2 \implies \ker d_1 = \text{im } d_2$.

Finally, $\ker \pi = (a, b) = \text{im } d_1$, even without regularness, and π is an epimorphism. $\square$

4.16

Let $\varphi : R^n \rightarrow R^m$ and $\psi : R^p \rightarrow R^q$ be two R -module homomorphisms, and let

$$\varphi \oplus \psi : R^n \oplus R^p \rightarrow R^m \oplus R^q$$

be the morphism induced on direct sums. Prove that

$$\text{coker}(\varphi \oplus \psi) = \text{coker } \varphi \oplus \text{coker } \psi.$$

Solution.

$$\begin{array}{ccccccc}
 R^n & \xrightarrow{\varphi} & R^m & \xrightarrow{\varphi'} & \text{coker } \varphi & & \\
 \downarrow i_1 & & \downarrow i'_1 & & \downarrow i''_1 & \searrow \sigma_1 & \\
 R^n \oplus R^p & \xrightarrow{\varphi \oplus \psi} & R^m \oplus R^q & \xrightarrow{\tau} & \text{coker } \varphi \oplus \text{coker } \psi & \xrightarrow{\lambda} & M \\
 \uparrow i_2 & & \uparrow i'_2 & & \uparrow i''_2 & \nearrow \sigma_2 & \\
 R^p & \xrightarrow{\psi} & R^q & \xrightarrow{\psi'} & \text{coker } \psi & &
 \end{array}$$

τ^* (curved arrow from $\text{coker } \psi$ to $R^m \oplus R^q$)
 τ (curved arrow from $R^m \oplus R^q$ to $\text{coker } \varphi \oplus \text{coker } \psi$)

Let $v \in R^n$, $w \in R^p$.

$$\tau \circ (\varphi \oplus \psi) \circ i_1(v) = i_1'' \circ \varphi' \circ \varphi(v) = i_1''(0) = (0, 0)$$

Similarly, $\tau \circ (\varphi \oplus \psi) \circ i_2 = 0$.

Hence, $\forall (v, w) \in R^n \oplus R^p$,

$$\begin{aligned} \tau \circ (\varphi \oplus \psi)(v, w) &= \tau \circ (\varphi \oplus \psi)(v, 0) + \tau \circ (\varphi \oplus \psi)(0, w) \\ &= \tau \circ (\varphi \oplus \psi) \circ i_1(v) + \tau \circ (\varphi \oplus \psi) \circ i_2(w) = (0, 0), \end{aligned}$$

so τ indeed annihilates $\varphi \oplus \psi$.

Moreover, if $\tau^* : R^m \oplus R^q \rightarrow M$ such that $\tau^* \circ (\varphi \oplus \psi) = 0$, then

$$\tau^* \circ i_2' \circ \psi = \tau^* \circ (\varphi \oplus \psi) \circ i_2 = 0$$

Since $\tau^* \circ i_2'$ annihilates ψ , it induces a unique $\sigma_2 : \text{coker } \psi \rightarrow M$ such that

$$\tau^* \circ i_2' = \sigma_2 \circ \psi'$$

Similarly, we obtain $\sigma_1 : \text{coker } \varphi \rightarrow M$ such that $\tau^* \circ i_1' = \sigma_1 \circ \varphi'$.

Together σ_1 and σ_2 induce $\lambda : \text{coker } \varphi \oplus \text{coker } \psi \rightarrow M$ such that

$$\sigma_1 = \lambda \circ i_1'', \quad \sigma_2 = \lambda \circ i_2''$$

We have then

$$\lambda \circ \tau \circ i_2' = \lambda \circ i_2'' \circ \psi' = \sigma_2 \circ \psi' = \tau^* \circ i_2'$$

and similarly $\lambda \circ \tau \circ i_1' = \tau^* \circ i_1'$.

It follows that $\forall (v, w) \in R^m \oplus R^q$,

$$\begin{aligned} \lambda \circ \tau(v, w) &= \lambda \circ \tau \circ i_1'(v) + \lambda \circ \tau \circ i_2'(w) \\ &= \tau^* \circ i_1'(v) + \tau^* \circ i_2'(w) \\ &= \tau^*(v, w) \implies \tau^* = \lambda \circ \tau \end{aligned}$$

If λ^* is another morphism such that $\tau^* = \lambda^* \circ \tau$, then $\sigma_2 \circ \psi' = \tau^* \circ i_2' = \lambda^* \circ \tau \circ i_2' = \lambda^* \circ i_2'' \circ \psi'$.

Since ψ' is an epimorphism, this would imply $\sigma_2 = \lambda^* \circ i_2''$. Similarly, we get $\sigma_1 = \lambda^* \circ i_1''$, and we must have $\lambda^* = \lambda$.

So, $\text{coker } \varphi \oplus \text{coker } \psi$ satisfies the universal property for $\text{coker}(\varphi \oplus \psi)$. □